Design Calculation Sheet (VBF-T5R1SINGLEMODEL-CALC-001)

Sheet: Input

Parameter	Formula / role	Value	Unit	Source
Base parameter set	anchor-table default (task names no electrode system)	OKane2022	-	entry 0 meta; base_params
Positive current collector thickness	mass lever (thinned 12->8 um)	8e-06	m	candidates/r4_archN_params.json
Negative current collector thickness	mass lever (thinned 8->6 um)	6e-06	m	candidates/r4_archN_params.json
Separator thickness	mass lever (default 25->10 um class)	1e-05	m	candidates/r4_archN_params.json
Positive electrode porosity	mass lever on measured capacity slack	0.4	-	candidates/r4_archN_params.json; R1 measured no capacity loss
Negative electrode porosity	transport lever	0.22	-	candidates/r4_archN_params.json
Electrolyte conductivity	transport bridge (plating suppression)	3.5	S/m	candidates/r4_archN_params.json
Electrolyte diffusivity	transport bridge	1.2e-09	m2/s	candidates/r4_archN_params.json
Cation transference number	transport bridge	0.55	-	candidates/r4_archN_params.json
Positive particle radius	kinetics (plating margin)	2e-06	m	candidates/r4_archN_params.json
Negative particle radius	kinetics (plating margin)	2.2e-06	m	candidates/r4_archN_params.json
Total heat transfer coefficient	cooling (safe window: h>45 overcools -> plating)	45	W/m2/K	candidates/r4_archN_params.json; R3 measured
Positive electrode thickness	base value	7.56e-05	m	OKane2022
Negative electrode thickness	base value	8.52e-05	m	OKane2022
Electrode area (height x width)	base value	0.1027	m2	OKane2022 Electrode height 0.065 x width 1.58

Sheet: CapacityEnergy

Item	Formula	Value	Unit	Source
1C discharge capacity (DFN)	simulation	5.0374	Ah	cell/r5_final_archN_1c_dfn.json:capacity_ah
1C discharge energy (DFN)	integral V*I_1C dt	18.37360559343354	Wh	cell/r5_final_archN_energy_dfn.json:energy_wh
Midpoint voltage (DFN)	voltage at mid discharge time	3.710580158893049	V	cell/r5_final_archN_energy_dfn.json:midpoint_voltage_v
SPMe cross-check ED	same caliber, SPMe 1C	531.85	Wh/kg	cell/r4_archN_energy.json (+0.25% vs DFN)
4C charge max temperature	4C_charge_45C DFN (T_amb 318.15 K)	330.5	K	cell/r4_archN_4c_dfn.json:T_max_K
4C charge anode potential min	min of anode_potential_v series	0.0215	V	cell/r4_archN_4c_dfn.json (plated=false)

Sheet: EnergyDensity

Item	Formula	Value	Unit	Source
Contract cell mass (electrolyte excluded)	sum over layers: th x (1-por) x density x area	34.4605754324	g	cell/r5_final_archN_energy_dfn.json:mass_kg
Gravimetric energy density	energy_wh / mass_kg	533.1775619787993	Wh/kg	cell/r5_final_archN_energy_dfn.json
Threshold check	criterion >= 500.94 Wh/kg	PASS (+32.2)	-	entry 0 criteria.stage2
Volumetric energy density	energy / (thickness x area)	968.1039210490741	Wh/L	cell/r5_final_archN_energy_dfn.json

Item	Formula	Value	Unit	Source
Layer stack thickness	sum of 5 layer thicknesses	184.8	um	cell/r5_final_archN_energy_dfn.json

Sheet: NPMass

Item	Formula	Value	Unit	Source
N/P ratio (voltage-cutoff equilibrium caliber)	OCP endpoint solve at 2.5/4.2 V with lithium conservation	1	-	_np_ratio.py (inferred; OKane2022 OCP CSV + concentrations)
Equilibrium capacity (N/P solve)	areal capacity x area	5.097	Ah	_np_ratio.py (simulated 1C: 5.037 Ah, +1.2% kinetic cutoffs)
Positive stoich window	x at 2.5 V -> 4.2 V	0.268 -> 0.851	-	_np_ratio.py
Negative stoich window	x at 4.2 V -> 2.5 V	0.905 -> 0.030	-	_np_ratio.py
Positive electrode mass	layer_kg_m2 x area	15.195935664	g	cell/r5_final_archN_energy_dfn.json:layer_kg_m2
Negative electrode mass	layer_kg_m2 x area	11.3090766984	g	same
Positive CC (Al) mass	layer_kg_m2 x area	2.21832	g	same
Negative CC (Cu) mass	layer_kg_m2 x area	5.521152000000001	g	same
Separator mass	layer_kg_m2 x area	0.2160910700000001	g	same
Negative-limited note	R2 measured: -8% neg thickness -> -7.9% capacity	neg binds capacity	-	log.jsonl R2 evaluate

Sheet: Process

Item	Formula	Value	Unit	Source
Positive areal density	th x (1-por) x density	147.96432	g/m2	calc-energy layer caliber
Negative areal density	th x (1-por) x density	110.117592	g/m2	calc-energy layer caliber
Positive compaction density	density x (1-por) / 1000	1.9572	g/cm3	OKane2022 density 3262 kg/m3
Negative compaction density	density x (1-por) / 1000	1.29246	g/cm3	OKane2022 density 1657 kg/m3
Electrolyte fill amount	pore volume x 1.2 g/cm3 x fill factor 1.0	6.616016160000001	g	pore volume from layer parameters; density literature value annotated
Formation recommendation	design recommended value	0.1C CC to 4.2 V, 25 C, 2 cycles	-	annotated: production-line value requires tuning